Introduction

Computer System Architecture.

Page No.
Date

characteristics of computer

(i) Automatic : works by itself without human intervention
(ii) speed : can perform calculations in few seconds.
(iii) Accuracy : output is always correct.
(iv) Diligence : is free from tiredness, lack of concentration.
(v) Versatility : can do many work at a time.
(vi) Power of remembering : remembers everything, large storage
(vii) NO I.Q. : no intelligence of its own } → disadvantages
(viii) No feelings : no feelings and no instincts.

Error can occur in computer due to human rather than technological weaknesses. We often refer to computer errors due to incorrect input data or unreliable programs as garbage-in-garbage-out. (GIGO)

Speed can be in seconds or milliseconds (10^{-3}), microseconds (10^{-6}), naroseconds (10^{-9}), picoseconds (10^{-12})
A powerful computer is capable of performing several billion (10^{9}) simple arithmetic operations per second.

Evolution of computers

(i) Blaise Pascal inverted first mechanical adding machine in 1642.
(ii) Baron Gottfried Wilhelm von Leibniz (Germany) inverted first calculator for multiplication in 1671.
(iii) keyboard machines originated in US around 1880.
(iv) Herman Hollerith came up with concept of punched cards in late 1970s.
(v) Charles Babbage, a nineteenth century Professor at Cambridge University, is considered the father of modern digital computer.

The Mark I computer (1937-44): also known as Automatic Sequence Controlled calculator, this was first fully automatic calculating machine designed by Howard A. Aiken (Harvard University) in collaboration with IBM (International Business Machine) It was capable of performing five basic arithmetic operations: addition, substraction, multiplication, division and table reference on number as big as 23 decimal digit.

The Atanasoff-Berry Computer (1939-42): Dr. John Atanasoff developed to solve certain mathematical equations. Also called ABC, after its inventor's name and his assistant, Clifford Berry. 45 vacuum tubes & capacitor

The ENIAC (1943-46): Electronic Numerical Integrator and Calculator (ENIAC) was first all electronic computer. Constructed at Moore school of Engineering (University of Pennsylvania, U.S.A) by J. Presper Eckert and John Mauchly. It was built because of military needs. 18,000 vacuum tubes.

The EDVAC (1946-52): Electronic Discrete Variable Automatic Computer. used the stored program concept in its design by Von Neumann. He also gave idea of storing both instructions and data in binary form. instead of decimal numbers or human readable words.

The EDSAC (1947-49): Britishers developed the Electronic Delay storage Automatic Calculator (EDSAC) by Professor Maurice Wilkes at Cambridge University Mathematical Laboratory.

Page No.
Date

The UNIVAC I (1951) : Universal Automatic Computer was first digital computer that was not "one of a kind". In 1952, IBM introduced IBM-701 commercial computer. UNIVAC marked the arrival of commercially available digital computers for business and scientific applications.

Data is raw material used as input to data processing.

Information is processed data obtained as output of data processing.

COMPUTER GENERATIONS

First Generation (1942 - 1955) (13 yr) - Vacuum tubes, electromagnetic relay memory, punched cards secondary storage.
Machine and assembly languages ; stored program concept, mostly scientific applications.
ENIAC, EDVAC, EDSAC, UNIVAC I, IBM-701

Second Generation (1955 - 1964) (9 yr) - transistors, magnetic core memory, magnetic tapes and disks secondary storage.
Batch operating system, high level language, scientific and commercial applications.
Honey Well 400, IBM 7030, CDC 1604, UNIVAC LARC

Third Generation (1964 - 1975) (11 yr) - ICs with SSI and MSI technologies (small scale integration, Medium scale integration), minicomputers.
Time sharing O.S., Standardization of high-level

programming language.
IBM 360/370, PDP-8, PDP-11, CDC 6600.

Fourth Generation (1975-1989) - ICs with VLSI technology (very large scale integration), microprocessor, semiconductor memory, PC (personal computers)
Operating system for PCs, GUI, UNIX OS, C, PC based applications.
IBM PC, Apple II, TRS-80.

Fifth Generation (1989- Present) - ICs with ULSI technology (Ultra large scale integration), notebook computers, powerful desktop PCs, workstation, very powerful mainframe, Internet.
WWW, multimedia, Internet based applications.
IBM notebooks, SUN workstation, PARAM 10000

Basic Computer Organization

Page No.
Date

Inputting: Process of entering data and instructions into a computer system.

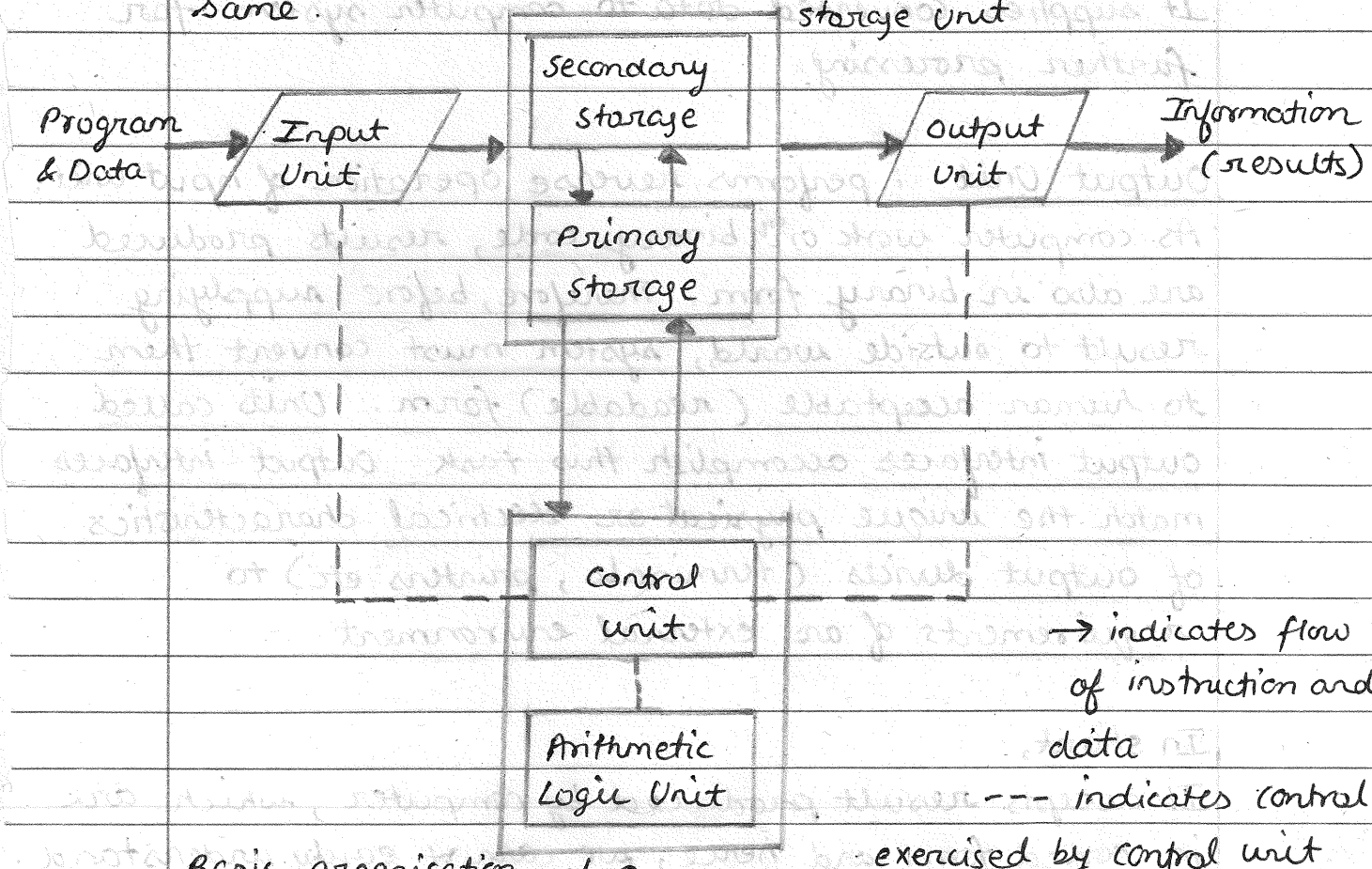
Storing: Saving data and instructions to make them readily available for initial or additional processing as and when required.

Processing: Performing arithmetic operations (add, substract, multiply, divide etc) or logical operations (comparisons like equal to, less than, greater than, etc) on data to convert them into useful informations.

Outputting: Process of producing useful information or results for user, such as printed report or visual display.

Controlling: Directing the manner and sequence in which above operation are performed.

Internal architecture of computer differs from one system modal to another. However, basic organization remains same.



Basic organisation of a computer system.

Page No.
Date

Input Unit

Data and instructions enter a computer through an input unit in form that depends upon the input device used.
Forex: data can be entered using keyboard in a manner similar to typing.
all input devices must transfer input signals to binary codes. Units called input interfaces accomplish this transformation. Input interfaces match the unique physical or electrical characteristics of input devices to requirements of computer system.

In short,
It accepts instructions and data from outside world
↓
It converts these data into computer acceptable form
↓
It supplies converted data to computer system for further processing.

Output Unit : performs reverse operation of input unit. As computer work with binary code, results produced are also in binary form. Therefore, before supplying result to outside world, system must convert them to human acceptable (readable) form. Units called output interfaces accomplish this task. Output interfaces match the unique physical or electrical characteristics of output devices (terminals, printers etc) to requirements of an external environment.

In short,
It accepts result produced by computer, which are in coded form and hence, we cannot easily understand them

Page No.
Date

↓

It converts these coded result to human acceptable (readable form)

↓

It supplies the converted result to outside world.

Storage Unit

Data entered into computer system through input units have to be stored inside computer before actual processing starts.

Similarly, results produced by a computer after processing have to be kept somewhere inside the computer system before being passed on to an input unit.

Moreover, computer must also preserve intermediate results for ongoing processing.

Storage unit provides space for storing data and instructions, intermediate results, results for output.

In short,

Data and instructions required for processing

↓

Intermediate results of processing

↓

Results for output, before they are released to an output device.

Page No.
Date

Storage unit of all computers is comprised of following two types of storage:

Primary storage: (main memory)
↳ used to hold pieces of program instructions and data, intermediate result of processing, and recently produced result on which computer is currently working. These pieces of information are represented electronically in main memory chip's circuitry and CPU can access it directly at a very fast speed.
Primary memory is volatile / temporary. ie as soon as computer switches off / resets, information is erased.
Primary memory is expersive (hence limited in size).
↳ is made up of semiconductor devices.

Secondary Storage: (auxiliary storage)
↳ is nonvolatile / permanent memory.
↳ is cheaper
↳ magnetic disk is most commonly used secondary storage medium. (like harddisk, floppy disks)
Secondary storage holds program instructions, data and information of those jobs on which computer system is not working.
Optical disks can also be used like CDs and CDROMs and magnetic tapes which were first form of secondary memory.

Central Processing Unit (CPU): CU + ALU
↳ also known as brain of computer.
↳ all major calculation and comparision
↳ responsible for activating and controlling operations of other unit of computer system.

Page No.
Date

Arithmetic Logic Unit (ALU)
↳ executions of instructions takes place in ALU during processing. (calculations and comparision).
↳ data and instructions stored in primary storage before processing are transferred as and when needed to ALU when processing takes place.
↳ Intermediate results generated in ALU are temporarily transferred back to primary storage until needed later.
↳ data can move from primary storage to ALU and back again to storage many times before processing overs.

Arithmetic operation → add, substract, multiply, divide
Logical operation → less than, equal to, greater than.

Control Unit
↳ acts as a central nervous system for other components of computer system.
↳ manages and coordinates entire computer system.
↳ it obtains instructions from program stored in main memory, interprets instructions, issues signals causing other units of system to execute them.

The system concept.
A system is a group of integrated parts that have a common purpose of achieving some objective(s). Hence, a system must have
(i) more than one element
(ii) all its element must be related logically.
(iii) all elements must be controlled in manner to achieve system goal.
A computer is often refered to as computer system because it is made up of integrated components (input unit, output unit, storage unit, CPU).

Page No.
Date

Some Input and output Devices

Input devices → Keyboard, Point and draw, Data scanning, Digitizer, Electronic card based devices, Speech recognition, vision based devices.

eg: Mouse, Keyboard, Joystick, Light pen, Graphics tablet, touchscreen, scanner, MICR, OCR, OMR, Web Camera, Card Reader, Track Ball

Output devices → Monitor, Printers, Plotters, Screen Image projector, Voice response system, speaker, headphones

for more read any book

like Pradeep K. Sinha and Preeti Sinha
Computer Fundamentals

Page No.
Date

Processor and Memory.

CPU (main memory / simple memory)

Central processing Unit (CPU)

Control Unit (CU) — Arithmetic Logic Unit (ALU)

Control Unit

↳ selects and interprets program instructions and then coordinates their execution.

↳ it has some special purpose registers and a decoder to perform these activities.

↳ The special purpose registers named instruction register and program control register holds the current instruction and next instruction to be executed and in this ways helps CU in instruction selection.

↳ Decoder has necessary circuitary to decode and interpret the meaning of every instruction supported by CPU. Each instruction is accompanied by microcode - very basic directions that tell CPU how to execute the instruction.

↳ called as central nervous system.

↳ it manages and coordinates the entire computer system its input and output units.

↳ it obtains instructions from program stored in main memory, interprets the instructions, and issues signals that causes other units of system to excute them.

Arithmetic Logic Unit (ALU)

↳ when control unit encounters an instruction that involves arithmetic operation or logic operation it passes control to ALU.

↳ ALU has some special purpose registers and necessary circuitry to carry out all arithmetic and

logic operations included in set of instructions supported by CPU

L For eg: control unit might load two numbers into ALU registers and then tell ALU to add them or check if they are equal.

Instruction Set

Every CPU has built in ability to execute a set of machine instruction called instruction set. Each processor has unique instruction set, machine language program written for one computer will generally not run on another computer with different CPU.

When a new CPU is developed it is ensured that its instruction set of its predecessor CPU plus some new ones. This manufacturing strategy is known as backward compatibility. and new CPU is said to be backward compatible with is predecessor. It allows user to upgrade their system easily without worrying about converting all their existing software.

Registers.

As instructions are interpreted and executed by a computer's CPU, there is a movement of information b/w various units of computer. In order to handle this and speed up rate of information transfer, a no. of special memory units called registers are used.

L these register hold information on temporary basis and are part of CPU (not main memory)

Page No.
Date

The length of register equals number of bits it can store.
eg: A register that can store 8 bit is called 8-bit register.
mostly CPU sold today have 32-bit or 64-bit registers. The length of registers of computer is sometimes called its word size. The bigger the word size, faster a computer can process a set of data.
eg: CPU with 32-bit registers can process data twice as large as one with 16-bit registers.

Although, no. of register varies from computer to computer, there are some registers common to all computerss.

(I) Memory Address Register (MAR) → it holds address of active memory location. It is loaded from program control register when an instruction is read from memory.

(II) Memory Buffer register (MBR) → It holds content of accessed (read/written) memory word. ie holds information on its way to and from memory. A instruction word placed in this register is transferred to instruction register.

(III) Program Control Register (PC) → it holds the address of next instruction to be executed. The instructions of program are stored in consecutive memory locations, and are executed in sequence, unless a branch instruction is encountered.

(N) Accumulator Register (A) → accumulates results and data to be operated upon / intermediate result. It is used during execution of most instructions. In many computers, there are more than one accumulator register.

(v) Instruction Register (I) → holds an instruction while it is being executed. ie current instruction being executed. As instruction is stored in register, operation part & address part of instruction is seperated. Address part is sent to MAR and operation part is sent to CU where it is decoded and interpreted.

(vi) Input / Output Register (I/O) → communicates with I/O devices. All input information such as instructions and data are transferred to this register by an input device, and all output information to be transferred to an output device.

A branch instruction is an operation that causes a transfer to non consecutive instruction.

Procesor Speed.

CU and ALU operations are usually synchronized by a built-in electronic clock (system clock) that emits millions of regularly spaced electric pulses per second. (clock cycles). Instructions are fetched, decoded and executed at proper intervals timed by specific number of clock cycles.

Shorter the clock cycle, faster the processor.

speed at which instruction is executed is related directly to computer's built-in clock speed, which is no of pulses per second.

clock speed is measured in megahertz (MHz) or gigahertz (GHz)

↓ (10^6)

↓ 10^9.

Page No.
Date

hertz means cycle per second.

→ major of computer have this speed.

CPU operating at 500 MHz can process data five times as fast as 100 MHz.

Types of Processor
- EPIC
- RISC
- CISC
- Multicore

(I) CISC Processor

earlier goals of CPU designers
↳ to provide more & more instruction in instruction set
↳ to optimize usage of expensive memory.

CPU with large instruction set, variable-length instructions, and a variety of addressing modes are said to employ CISC (Complex Instruction set computer) architecture since CISC processor possess so many processing features, they make job of machine language programmers easier.

However, they are complex and expensive to produce.

Most PCs today use CISC processor.

(II) RISC processor.

Some CPU designers realised that many instructions supported by CISC based CPU are rarely used.

CPU with small instruction set, fixed-length instructions, and reduced references to memory to retrieve operands are said to employ RISC. (reduced instruction set computer) architecture.

However, they are simple, fast and inexpensive (less expensive)

are designed.

In RISC, all instructions are such a way that they retrieve operands stored in registers in CPU rather than memory.

EPIC processor

Explicitly parallel instruction computing (EPIC) technology. It allows software to communicate explicitly to processor when operations can be done in parallel. For this it uses coupling b/w compiler and processor.

Processor based on EPIC architecture are simpler and more powerful than traditional CISC or RISC processor.

These processor are mainly targeted to next generation, 64-bit, high-end server and work-station market. (not fo PCs)

Multicore Processors.

Till recently, approach used for building faster processor was to keep reducing size of chips while increasing no. of transistors they contain.

In multicore processor, each core has its independent cache, thus providing OS with sufficient resources to handle multiple applications in parallel. When a single core chip runs multiple programs, OS assigns a time slice to work on one program and different time slices for other programs. This can cause conflict, error or slowdown. However, multiprograms can run at same time on multicore chip with each

core handling a separate program.

Multicore advantages over single core processor.

(I) They enable building of computers with better overall system performance by handling more work in parallel.

(II) Consumes less power and generate less heat. Also called energy-efficient or power-aware processor technology.

(III) they can share architectural components, such as memory elements, memory management.

(IV) signaling b/w cores can be faster and use less electricity.

Multicore limitations:

(I) applications must be redesigned so that processor can run them as multiple threads.

(II) programmers must find good places to break up applications, divide the work into roughly equal pieces and determine best times for thread to communicate with one another.

(III) software vendors often charge customers for each processor that will run software.

Chip makers like Intel, IBM and Sun have already introduced multicore chips like dual core (2 core per chip), quad-core (4 core per chip), 8 cores per chip and 16 cores per chip.

Page No.
Date

Main memory.

CPU can process data at rate of 5 nanoseconds/byte and disk reader can read data at a speed of about 5 microsecond/byte. This leads to slow performance. Main memory is temporary storage area built into the computer hardware. Instructions and data of program reside mainly in this area where CPU is executing program.

This memory consist of some IC chips either on motherboard or on a small circuit board attached to motherboard of computer system. This allows CPU to store and retrieve data very quickly. The rate of fetching data from memory is of order 50 nanoseconds/byte. Here, rate of data fetching from main memory is about 100 times faster than that of from high-speed secondary storage like disk.

Storage Evaluation criteria.

(i) storage capacity : storage capacity is less than secondary storage

(ii) faster access time

(iii) lower cost per bit of storage is required (refers to cost of storage unit for given storage capacity). Primary units have higher cost per bit storage

(iv) Volatile (temporary) : Yes, they are volatile.

(v) they have random access property and secondary storage units have pseudo-random access or sequential access.

If time taken to access a piece of data from storage unit is independent of location of data in storage unit, it is called random access storage/memory.

Page No.
Date

Main memory Organization.

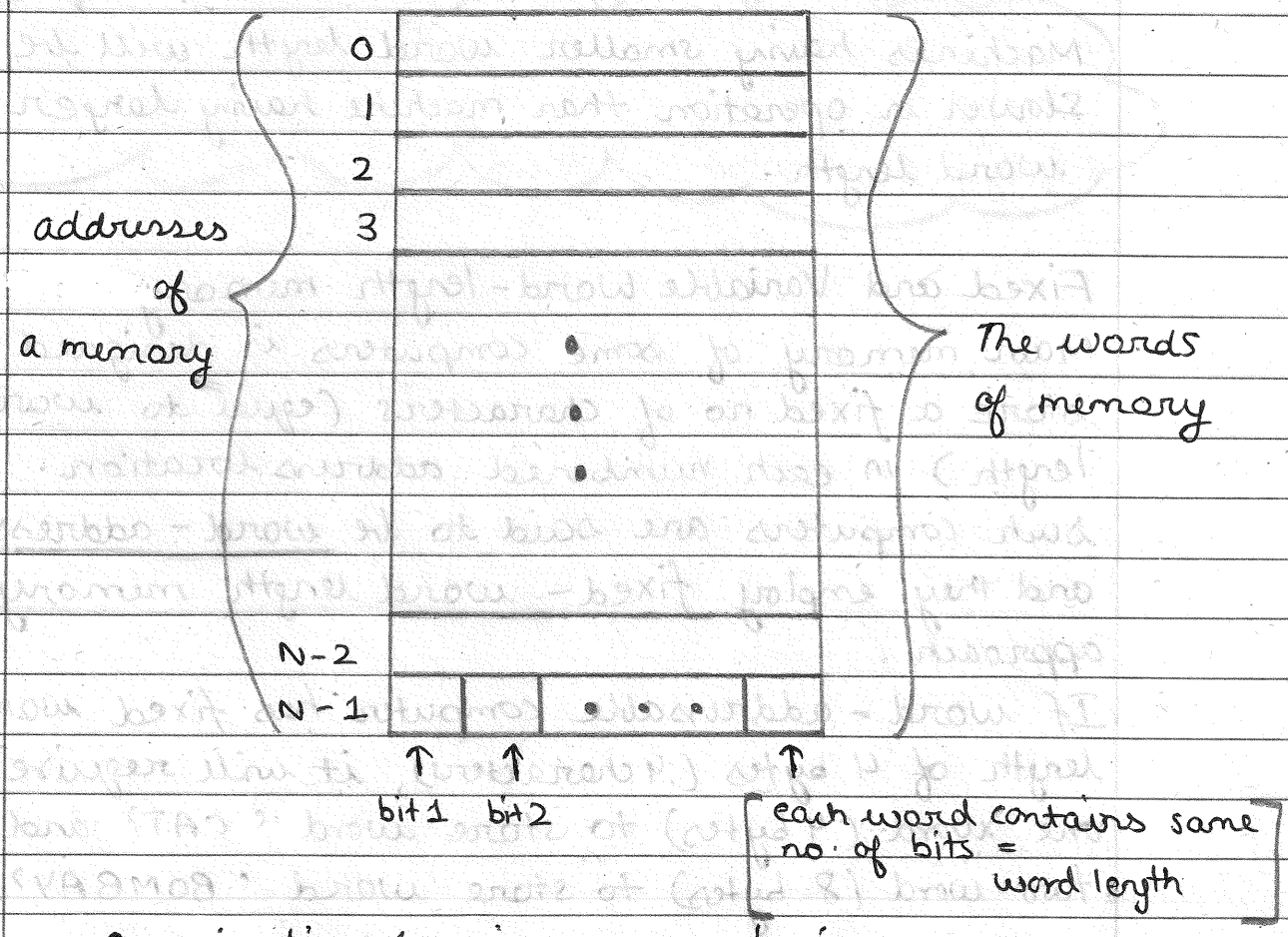
Main memory is made up of several small storage areas called locations or cells.

Each of these location can store a fixed no. of bits called word length of memory.

A given memory is divided into N words, where generally N is some power of 2. Each word or location has a built-in and unique no. assigned to it. This no is called address of location and is used to identify location.

The first address starts at 0 and highest address equals no. of words that can be stored in memory − 1 (minus 1)

ex: If memory has 1024 locations, address rayes b/w 0 to 1023.



Organization of main memory having N words.

Page No.
Date

Why More Bits?

You have heard about 8-bit, 16-bit, 32-bit computer etc. They refer to word length of memory of computer in terms of its total no. of bits per memory word.

What is advantage of having more no of bits per word, instead of having more words of smaller size (length)?

Ans: Imagine a highway with eight lanes and heavy traffic and is expanded to eight lanes.. 'Lanes' refers to 'Bits' on microchip. More bits mean a more rapid flow of electronic signals, in other words faster computer.

(Retrieving) Read is non-destructive operation, write is destructive operation (entering data).

Machines having smaller word length will be slower in operation than machine having larger word length.

Fixed and Variable Word-length memory.

Main memory of some computers is designed to store a fixed no. of characters (equal to word length) in each numbered address location.

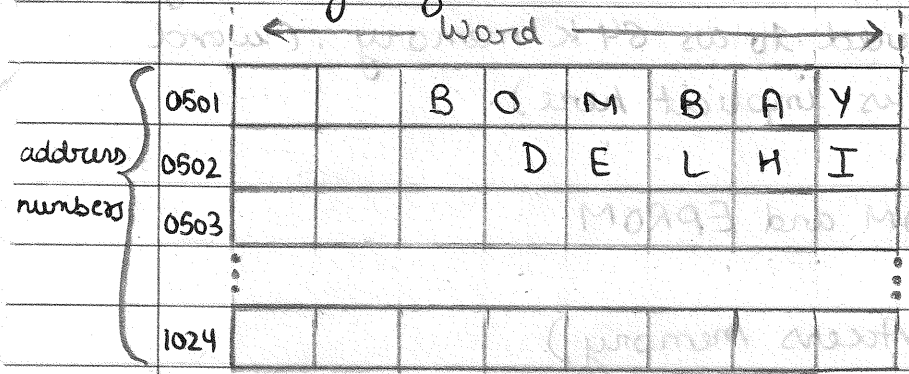
Such computers are said to be word-addressable and they employ fixed-word length memory approach.

If word-addressable computer has fixed word length of 4 bytes (4 characters), it will require one word (4 bytes) to store word 'CAT' and two word (8 bytes) to store word 'BOMBAY'.

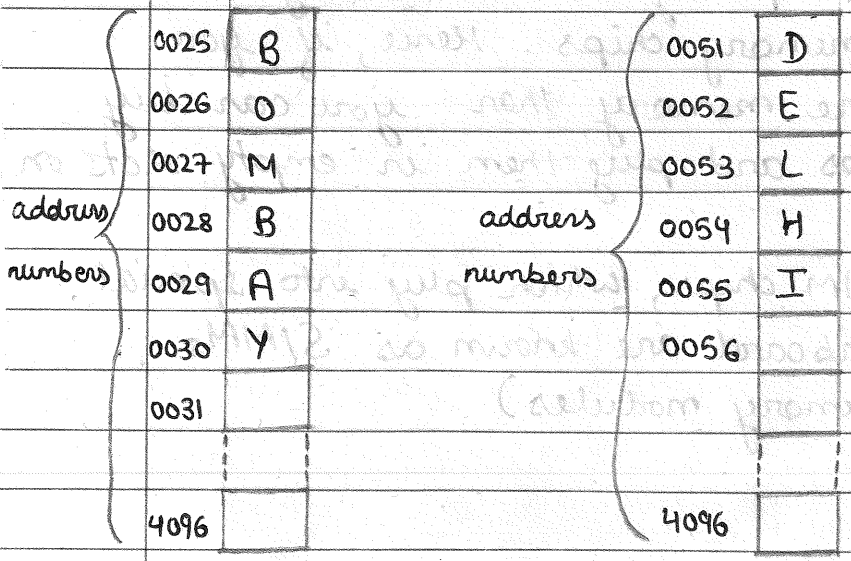
In many computers, main memory is designed to store a single character at each numbered address. These computer are called character-addressable and they employ a variable-word length memory approach.
Hence, only 3 bytes will be required to store 'CAT' and 6 bytes for 'BOMBAY'.

Word addressable computers posess faster calculating capabilities, used for gaining speed of calculation.
Character addressable (variable length) is used for optimizing use of storage space.

Most modern computers employ fixed-word length memory organisation.



(a) Fixed-word-length memory.



(b) variable-word-length memory

Page No.
Date

Main memory capacity

↳ normally stated in terms of kilobytes (kB), which is equal to 1024 (2^{10}) bytes of storage, megabytes (MB) = 2^{20} bytes, gigabytes (GB) = 2^{30} bytes.

ex: A computer having 32 MB of memory is capable of storing 32×2^{20} bytes or characters.

A memory with 4096 locations with each location storing 16 bits is called "16-bit, 4096-word memory" or "4 k, 16-bit memory".
Similarly, a memory having 2^{15} words with each word of 16 bits is called "32 k, 16 bit memory".

A memory having 2^{16} words with each word of 8 bits is referred to as 64 k memory. (word size of 8-bits is implicit here).

RAM, ROM, PROM and EPROM

RAM (Random Access memory)

A computer motherboard is designed in a manner that its memory capacity can be easily enhanced by adding more memory chips. Here, if you need to have more memory than you can buy more memory chips and plug them in empty slots on motherboard.
The additional RAM chips, which plug into special sockets on motherboard are known as SIMMs.
(single in-line memory modules)

Page No. Date

↳ also known as field stores / dead stores

ROM (Read only memory)

↳ non-volatile (permanent), cannot be altered by usual programs.

↳ Storing data permanently into this kind of memory is called "burning in the data" because data is stored by using fuse-links. once a fuse link is burnt, it is permanent.

↳ Instead of building electronic circuits for these operations, special programs are written to perform them. (these are called micro programs because they deal with low-level machine functions) and are essential substitute for additional hardware.

ROMs are used to store these micro-programs so that they cannot be modified by user.

ex: System boot program is a micro-program. Contains set of start up instructions to check if system hardware like memory, I/O devices, etc are functioning properly.

Note:

Microprograms are used everytime computer is switched on and needs to be retained when computer is switched off.

PROM (Programmable Read only memory)

There are two types of (ROM) – manufacturer-programmed and user programmed.

A manufatured programmed ROM is one in which data is burnt in by manufacturer of electronic equipment in which it is used.

eg: manufacturer may store system boot program

permanently in ROM chip located on motherboard of each PC manufactured by it.

A user programmed ROM is one in which a user can load and store "read only" programs and data. It is possible for user to "customize" a system by converting his/her programs to micro-programs and storing them in user programmed ROM chip. Such ROM is called "Programmable Read-only Memory" (PROM) because a user can program it. once a chip has been programmed, PROM becomes ROM.

Erasable Programmable Read only Memory (EPROM)
It is possible to erase information stored in an EPROM chip and chip can be reprogrammed to store new information.
EPROM are often used as R&D personnel who frequently change microprograms to test efficiency of computer system with new programs.

EPROM chips are of two types
(i) erased by ultraviolet light (called UVEPROM)
(ii) erased by high voltage electric pulses (called EEPROM) also called flash memory.
It is easier to alter EEPROM chip than UVEPROM)

EEPROM is called flash memory because of ease with which programs stored init can be altered.
flash memory is used in many new I/O and storage devices like USB and MP3 music player.

EEPROM - electrically PROM UVEPROM - ultra violet PROM